DIGITAL EGYPT PIONEERS INITIATIVE / DEVOPS TRACK

AdaptiveOps — Automated E-Commerce Deployment Platform

Literature Review and Lecturer Evaluation | Project 4 | Version 1.4 | 15 October 2026

PROJECT	AdaptiveOps | Project 4: Automated E-Commerce Deployment Platform
REVIEW BASIS	Official DEPI brief, GitHub upstream documentation and official technical guides
LECTURER STATUS	Project 4, its approved scope and use of an existing application are confirmed by the team following lecturer review. Detailed grading rubric not supplied.
DATE	15 October 2026

1. Purpose and interpretation of DEPI instructions

The updated DEPI guidelines call this section "Literature Review" and request lecturer feedback, improvements and grading criteria; earlier guidance uses "Lecturer Review". This report combines technical references and confirmed lecturer decisions. No numerical grading rubric was supplied.

2. Problem and technical background

Multi-service releases depend on configuration, networking, data stores and runtime health. Versioned containers, scripted deployments, CI/CD and monitoring support repeatable releases. The planned AdaptiveOps implementation includes measured scaling and explicit rollback with functional recovery checks.

3. Existing application approaches considered

Application / source	Documented strengths	Fit / limitation for DEPI
Google Online Boutique [R1]	11 application microservices plus Redis; web, SearchProducts API, mock payment and cart. Kubernetes workload [R1, R7].	SELECTED. Local run pending. Search API exists; no search route found in inspected frontend. Jenkins/AWS/rollback integration remains team work.
AWS Containers Retail Sample [R2]	UI, catalog, cart, orders, checkout; Docker Compose/Kubernetes; AWS infrastructure examples; selectable data backends	Alternative if local feasibility fails. Separate mock payment functionality not verified in initial review; stronger AWS-specific examples.

Google Online Boutique is the selected workload. At inspected upstream commit 38e7348eb289, SearchProducts uses gRPC; no dedicated frontend search route was found, and Redis uses emptyDir [R7]. This configuration does not ensure cart persistence across Pod replacement or provide relational orders. These limitations will be tested and documented. The referenced commit is not yet the team's pinned deployment version.

4. Related engineering practices and gaps

Practice	Reference insight	Work the team must produce
Kubernetes HPA [R3]	Scaling depends on observed workload metrics and the required metrics APIs / resource requests.	Set valid HPA targets; generate appropriate load; record replica and resource metrics.
Prometheus + Grafana [R4]	Prometheus stores queryable metrics; Grafana uses its Prometheus data source to display them.	Choose verified endpoints/exporters and create service/system dashboards.
Automated release recovery [R8]	A failed Deployment is reported; rollback requires an explicit action. Upstream /_healthz alone does not verify dependencies [R7].	Check rollout and functional paths; retain known-good image/config; invoke rollback and verify recovery.
Infrastructure-as-Code	Repeatable host/resource configuration can reduce manual setup differences.	Ansible playbooks and Terraform scripts for an affordable, permitted AWS subset.

5. Lecturer feedback & evaluation record

Review topic	Approved decision / documented information
Project selection	Project 4 - Automated E-Commerce Deployment Platform - approved by the lecturer.
Application reuse	Use of an existing open-source e-commerce application is approved.
AWS scope	Implementation of feasible and available AWS services is approved. Terraform scripts remain a project deliverable; cloud execution is subject to account limits and cost.
Detailed review comments	Project approach confirmed. No separate itemized written review comments supplied in the source materials.
Detailed grading rubric	A detailed numerical grading rubric was not included in the available DEPI documents.
Open clarifications	Validate and document the search interface and Redis data-store behavior during implementation; report any limitations in the technical review.

6. Proposed technical improvements (project team recommendations)

- Pin a released Online Boutique commit or tag, preserve its Apache-2.0 license and document exactly which files come from the original project.
- Keep the initial deployment local and independently demoable; evaluate EC2/S3 only once AWS permission, cost and cleanup are understood.
- Test web/cart/mock checkout and the catalog search API; document Redis persistence limits and actual HPA/rollback behavior.
- Capture Prometheus targets, Grafana panels and actual Jenkins logs rather than describing theoretical capability.
- Use cross-tool technical walkthroughs so all six DevOps trainees can explain the deployed system and their contributions.

7. References (official project and technical sources)

- [R1] GoogleCloudPlatform, Online Boutique README and license: https://github.com/GoogleCloudPlatform/microservices-demo (accessed 15 October 2026).
- [R2] AWS Containers, Retail Store Sample App, project README and deployment options: https://github.com/aws-containers/retail-store-sample-app (accessed 15 October 2026).
- [R3] Kubernetes Documentation, Horizontal Pod Autoscaling: https://kubernetes.io/docs/concepts/workloads/autoscaling/horizontal-pod-autoscale/ (accessed 15 October 2026).
- [R4] Prometheus Documentation, Grafana support for Prometheus: https://prometheus.io/docs/visualization/grafana/ (accessed 15 October 2026).
- [R5] DEPI, R5 Project Documentation Last Updates (uploaded official guidance), pp. 1-2.
- [R6] DEPI, R5 Projects Instructions - Final for Trainees (uploaded official guidance), pp. 1-3.
- [R7] Google source at commit 38e7348eb289eb5b87c0c6e8cb19ced0449dc389: src/frontend/main.go; src/productcatalogservice/product_catalog.go; protos/demo.proto; kubernetes-manifests/cartservice.yaml. https://github.com/GoogleCloudPlatform/microservices-demo/tree/38e7348eb289eb5b87c0c6e8cb19ced0449dc389 (inspected 15 October 2026).
- [R8] Kubernetes, Deployments, failed progress and rollback: https://kubernetes.io/docs/concepts/workloads/controllers/deployment/ (accessed 15 October 2026).

Document status: project scope and team responsibilities confirmed. Upstream references reviewed; deployment, performance, autoscaling and rollback have not yet been validated. GitHub submission is subject to separate verification.